Magic squares as orthogonality conditions: the permutation moment threshold and random line-sum energies

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Abstract

We regard the centred entries of an $n \times n$ arrangement of $1, \dots, n^2$ as a function z on the centred grid; the semi-magic and magic conditions say that z is orthogonal to the centred row, column and diagonal indicators. Replacing the line indicators by the powers x^m, y^m $(m \le k)$ of the column and row coordinates gives the degree-k moment conditions, which for real-valued z are equivalent to equal row and column sums exactly when $k \ge n-1$. The permutation threshold $k^*(n)$ is the least k for which the degree-k conditions force every arrangement to be semi-magic. These conditions say that the column deviations are the coefficients of an integer polynomial divisible by $(t-1)^{k+1}$, which connects $k^*(n)$ with the Prouhet–Tarry–Escott problem and with the Littlewood-type problems of Borwein, Erdélyi and Kós. We prove $(\sqrt{2}-o(1))\sqrt{n} \le k^*(n) \le \sqrt{\pi e\,(n-1)\log(n^4/4)}$, the lower bound by a box-principle construction inside semi-magic squares with many complementary pairs of rows, and we determine $k^*(n)$ for $n \le 40$: $k^*(n) = n-1$ for $n \le 13$, $k^*(n) = n-2$ for $14 \le n \le 17$, and $k^*(n) \le n-3$ for every $n \ge 18$. For a uniformly random arrangement we prove an exact formula for the variance of a quadratic form with zero row sums in the centred entries. It gives closed forms for the mean and variance of the sum of squared line-sum deviations and of a four-term low-order energy; the latter, divided by its mean, converges in distribution to Exp(1). We also give the null distribution of Loly's moment-of-inertia invariant, and the exact number 14,639,440 of 4×4 arrangements satisfying the four low-order conditions, of which 7,040 are magic.

Keywords: magic squares, Prouhet–Tarry–Escott problem, polynomials with integer coefficients, permutation statistics, combinatorial central limit theorem, moment of inertia
MSC 2020: 05B20 (primary); 05A05, 11C08, 60C05, 60F05

1 Introduction

Magic squares are among the oldest objects of recreational mathematics: the 3×3 Lo Shu appears in Chinese sources more than two millennia old, a 4×4 square is engraved in Dürer's *Melencolia I* (1514), and constructions for every order have been known since the seventeenth century [3, 28]. The modern literature treats them through enumeration and the geometry of the associated polytopes [32, 12, 1, 5, 4], through spectral and matrix properties [21, 26, 27, 9], and, in work of Loly and collaborators, through physical invariants such as the moment of inertia and the multipole moments [19, 30, 31].

Definition 1.1 (Magic square)**.** An *arrangement* of order n is a bijection $S = (s_{ij})$ from the cells of the $n \times n$ grid onto $\{1, \dots, n^2\}$. It is a *magic square* if every row, every column and both main diagonals sum to the magic constant $M(n) = n(n^2+1)/2$, and *semi-magic* if the rows and columns do.

We centre both the grid and the entries (Definition 2.1): the cell (i, j) sits at $(x_j, y_i) = (j - \frac{n-1}{2}, \frac{n-1}{2} - i)$ and carries the value $z_{ij} = s_{ij} - \frac{n^2+1}{2}$. The condition "line L sums to $M(n)$" then says that z is orthogonal to the centred indicator of L, so an arrangement is magic if and only if z is orthogonal to a fixed $2n$-dimensional space W of test functions. The dimension count is the classical fact that the real magic squares of order $n \geq 3$ form a space of dimension $n^2 - 2n$ [37], [33, §4.6], and the squared length of the vector of line deviations, $\sum_L \delta_L^2$, is the energy of the Monte Carlo study of magic squares by Pinn and Wieczerkowski [29]. This reformulation is elementary; we use it to pose the questions treated below. The convex hull of the points (x_j, y_i, z_{ij}) is studied in the companion paper [23] and is not used here.

The permutation threshold. Replacing the line indicators by the powers x^m and y^m, $1 \leq m \leq k$, gives the *degree-k conditions* $\sum_{ij} z_{ij} x_j^m = \sum_{ij} z_{ij} y_i^m = 0$. They hold for every semi-magic square, and for real-valued z they imply equal row and column sums if and only if $k \geq n - 1$ (Vandermonde). For arrangements, integrality and the bounded size of the line sums lower this threshold. Let $k^*(n)$ be the least k such that every arrangement satisfying the degree-k conditions is semi-magic. The degree-k conditions say that the polynomial $\sum_j \gamma_j t^j$ formed from the column deviations is divisible by $(t-1)^{k+1}$, and similarly for the rows, so that the positive and negative parts of γ form a solution of a Prouhet–Tarry–Escott system [38, 7]. Our main result (Theorem 3.7) is

$$(\sqrt{2} - o(1))\sqrt{n} \;\leq\; k^*(n) \;\leq\; \sqrt{\pi e\,(n-1)\log(n^4/4)} \;\leq\; 2\sqrt{\pi e}\,\sqrt{n \log n},$$

together with the exact values $k^*(n) = n - 1$ for $3 \leq n \leq 13$, $k^*(n) = n - 2$ for $14 \leq n \leq 17$, the bound $k^*(n) \leq n - 3$ for all $n \geq 18$, and the exact values for $n \leq 40$ (Table 1). The upper bound is a Jensen-formula argument of the type used by Borwein, Erdélyi and Kós for polynomials with bounded coefficients [8]. The lower bound needs the converse direction, namely that suitable integer vectors are column-deviation vectors of arrangements whose row sums all equal $M(n)$; we obtain it from semi-magic squares with many pairs of complementary rows, in which swaps realise prescribed column deviations, combined with a box-principle argument. Whether $k^*(n)$ grows like $\sqrt{n}$ or like $\sqrt{n \log n}$ remains open.

Random arrangements. For a uniformly random arrangement each $\mathrm{Cov}(\varphi, Z)$ is a linear permutation statistic, so the energy $E_{\mathrm{full}} = N^{-2} \sum_L \delta_L^2$ (all $2n + 2$ lines) and the low-order energy E_{low}, which combines the degree-1 and the two diagonal conditions (Definition 2.4), are quadratic forms in a random permutation of the centred values. We prove a closed formula for the variance of such a form in terms of three invariants of its matrix (Theorem 4.3), a special case of the classical moment formulas for quadratic assignment statistics [22, 16], and obtain the exact means and variances of both energies for every n (Theorem 4.4). By Hoeffding's combinatorial central limit theorem, $E_{\mathrm{low}} / \mathbb{E}[E_{\mathrm{low}}] \to \mathrm{Exp}(1)$ (Corollary 4.5). The same tools give the null distribution of Loly's moment of inertia $I(S) = \sum s_{ij}(x_j^2 + y_i^2)$ (Corollary 4.2): magic squares sit exactly at its mean.

The low-order class for $n = 4$. Exactly 14,639,440 of the 16! arrangements of order 4 satisfy the four conditions $E_{\mathrm{low}} = 0$, and 7,040 of them are magic (Proposition 5.1); the count is 0.80 times a Gaussian local-limit estimate.

Classical and new material. Orthogonality to the line space, the Vandermonde threshold, the vanishing of the separable moments of a semi-magic square and Loly's invariant [19] are classical or elementary, and the single-swap formula of Remark 5.2 is the incremental evaluation used in local search. To our knowledge Theorem 3.7, the closed forms of Theorem 4.4 with the limit law, Corollary 4.2 and the count in Proposition 5.1 have not appeared before.

Organisation. Section 2 fixes notation and collects the tools. Section 3 contains the moment hierarchy and the permutation threshold. Section 4 treats random arrangements, Section 5 the computations for small n, and Section 6 lists open questions. The appendices contain the computational methods, the construction of the squares used for the lower bound, and the derivation of the variance formulas.

2 Notation and tools

2.1 Centred coordinates

Throughout, $n \geq 3$ and $N = n^2$. Cells are pairs $c = (i, j)$ with $0 \leq i, j \leq n-1$ (i the row from the top, j the column from the left).

Definition 2.1 (Centred coordinates)**.** To the cell (i, j) of an arrangement S assign

$$x_{ij} = x_j = j - \frac{n-1}{2}, \qquad y_{ij} = y_i = \frac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \frac{N+1}{2}. \tag{1}$$

We call $z = z(S)$ the *centred value function* of S, and write X, Y, Z for the functions $c \mapsto x_c, y_c, z_c$.

The function z is the vertical coordinate of the points (x_c, y_c, z_c) studied in [23]; we do not call it a height function, a term that in combinatorics refers to tilings [34]. A *line* is one of the n rows, the n columns, the main diagonal $\{i = j\}$ or the anti-diagonal $\{i + j = n - 1\}$. Each line has n cells, and since $M(n) = n \cdot \frac{N+1}{2}$ the *deviation* of a line is

$$\delta_L(S) = \sum_{c \in L} s_c - M(n) = \sum_{c \in L} z_c. \tag{2}$$

We write ρ_i, γ_j, δ_{m}, δ_{a} for the deviations of row i, column j and the two diagonals; $\sum_i \rho_i = \sum_j \gamma_j = 0$. On the space $\mathcal{F} = \mathbb{R}^{\mathrm{cells}}$ we use $\langle f, g \rangle = \frac{1}{N} \sum_c f_c g_c$, and $\mathrm{Cov}(\varphi, f) = \langle \varphi - \bar{\varphi}\mathbf{1}, f - \bar{f}\mathbf{1} \rangle$ is the empirical covariance over the N cells. R_i, C_j, D_{m}, D_{a} denote the line indicators.

Lemma 2.2. *For every arrangement, $\bar{x} = \bar{y} = \bar{z} = 0$, so $\mathrm{Cov}(\varphi, Z) = \langle \varphi, z \rangle$ for every $\varphi \in \mathcal{F}$; $\mathrm{Var}(X) = \mathrm{Var}(Y) = \frac{n^2-1}{12}$, $\mathrm{Var}(Z) = \frac{N^2-1}{12}$, $\mathrm{Cov}(X, Y) = 0$; and $\mathrm{Cov}(\mathbf{1}_L, Z) = \delta_L / N$ for every line L.*

Proof. X, Y, Z are uniform on n, n and N equally spaced points centred at 0, and X, Y are independent under the uniform distribution on cells; the last claim is (2). □

2.2 The line space and the energies

Let $\tilde{\mathbf{1}}_L = \mathbf{1}_L - \frac{1}{n}\mathbf{1}$ be the centred indicator of a line and $W_R = \mathrm{span}\{\tilde{\mathbf{1}}_{R_i}\}$, $W_C = \mathrm{span}\{\tilde{\mathbf{1}}_{C_j}\}$, $W_D = \mathrm{span}\{\tilde{\mathbf{1}}_{D_{\mathrm{m}}}, \tilde{\mathbf{1}}_{D_{\mathrm{a}}}\}$, $W = W_R + W_C + W_D \subset \mathbf{1}^{\perp}$.

Lemma 2.3 (The line space)**.** *Let $n \geq 3$. For lines L, L', $N\langle \tilde{\mathbf{1}}_L, \tilde{\mathbf{1}}_{L'} \rangle = |L \cap L'| - 1$. Consequently:*

1. *W_R consists of the zero-mean functions of the row index and W_C of the zero-mean functions of the column index; both have dimension $n-1$, and the orthogonal projection of z onto them is $(i, j) \mapsto \rho_i / n$, resp. γ_j / n.*
2. *W_R, W_C, W_D are mutually orthogonal, and the Gram matrix of $(\tilde{\mathbf{1}}_{D_{\mathrm{m}}}, \tilde{\mathbf{1}}_{D_{\mathrm{a}}})$ is Γ_D / N with $\Gamma_D = \left(\begin{smallmatrix} n-1 & \epsilon_n \\ \epsilon_n & n-1 \end{smallmatrix}\right)$, $\epsilon_n = 0$ for odd and -1 for even n; its eigenvalues are $n-1, n-1$ (odd n) and $n-2, n$ (even n).*
3. *$W = W_R \oplus W_C \oplus W_D$ has dimension $2n$, and $z \perp W$ if and only if S is magic.*

Proof. Each line has n of the N cells, which gives the inner products: a row meets every column and each diagonal in one cell, two distinct rows (or columns) are disjoint, and the diagonals meet in the centre cell if and only if n is odd. The projection onto the functions of the row index replaces z by its row means ρ_i/n. Finally $\langle \tilde{\mathbf{1}}_L, z\rangle = \delta_L/N$ by Lemma 2.2. □

Part 3 is equivalent to the classical fact that the real magic squares of order $n \geq 3$ (all line sums equal to a common, arbitrary value) form a space of dimension $N - 2n$ [37], [33, §4.6].

Definition 2.4 (Energies)**.** For an arrangement S with centred value function z,

$$E_{\text{full}}(S) = \sum_i \operatorname{Cov}(R_i, Z)^2 + \sum_j \operatorname{Cov}(C_j, Z)^2 + \operatorname{Cov}(D_{\text{m}}, Z)^2 + \operatorname{Cov}(D_{\text{a}}, Z)^2, \tag{3}$$

$$E_{\text{low}}(S) = \operatorname{Cov}(X, Z)^2 + \operatorname{Cov}(Y, Z)^2 + \operatorname{Cov}(D_{\text{m}}, Z)^2 + \operatorname{Cov}(D_{\text{a}}, Z)^2. \tag{4}$$

By Lemma 2.2,

$$N^2 E_{\text{full}}(S) = \sum_L \delta_L(S)^2 = \sum_i \rho_i^2 + \sum_j \gamma_j^2 + \delta_{\text{m}}^2 + \delta_{\text{a}}^2, \tag{5}$$

the sum of the squared deviations of all $2n + 2$ lines, a non-negative integer. This is the energy of the Monte Carlo study of Pinn and Wieczerkowski [29], and we follow them in calling E_{full} and E_{low} energies. $E_{\text{full}}(S) = 0$ if and only if S is magic, and E_{full} depends on z only through its projection onto W (Figure 1). Both energies are invariant under the dihedral group D_4 of the square, which permutes the lines and maps $\{\pm X, \pm Y\}$ to itself. All n rows and n columns are kept in (3): the last row and column are redundant for the zero set, but omitting them would destroy the D_4-invariance.

Remark 2.5 (The four low-order conditions). $E_{\text{low}} = 0$ consists of four linear conditions on z, and "magic" of $2n$ independent ones; for every $n \geq 3$ the four conditions are therefore strictly weaker than the magic conditions on real-valued z. For permutations the two coincide when $n = 3$ (Proposition 3.14) but not when $n = 4$: the arrangement

$$S = \begin{pmatrix} 9 & 4 & 12 & 3 \\ 16 & 1 & 14 & 7 \\ 10 & 15 & 11 & 8 \\ 2 & 5 & 6 & 13 \end{pmatrix}$$

has both diagonal sums 34, row sums $(28, 38, 44, 26)$ and column sums $(37, 25, 43, 31)$, and $\operatorname{Cov}(X, Z) = \operatorname{Cov}(Y, Z) = 0$; so $E_{\text{low}}(S) = 0$ while $N^2 E_{\text{full}}(S) = 396$. Proposition 5.1 counts all such arrangements.

2.3 Permutation statistics

Let π be a uniformly random permutation of $\{1, \dots, N\}$ and $a, a', b \in \mathbb{R}^N$. For the linear permutation statistics $\Phi_a = \sum_i a_i b_{\pi(i)}$ one has $\mathbb{E}\Phi_a = N\bar{a}\bar{b}$ and, by the theorem of Wald and Wolfowitz [36, 25] (polarised),

$$\operatorname{Cov}(\Phi_a, \Phi_{a'}) = \frac{1}{N-1} \sum_i (a_i - \bar{a})(a'_i - \bar{a}') \sum_j (b_j - \bar{b})^2. \tag{6}$$

For a uniformly random arrangement, $z_c = b_{\pi(c)}$ with b the centred values $k - \frac{N+1}{2}$; we put

$$\sigma_n^2 = \frac{\operatorname{Var}(Z)}{N-1} = \frac{N+1}{12} = \frac{n^2+1}{12}. \tag{7}$$

Hoeffding's combinatorial central limit theorem [15] concerns $\sum_i c_N(i, \pi(i))$ for an $N \times N$ array c_N. Let $d_N(i,j) = c_N(i,j) - \frac{1}{N}\sum_g c_N(g,j) - \frac{1}{N}\sum_h c_N(i,h) + \frac{1}{N^2}\sum_{g,h} c_N(g,h)$ be the doubly

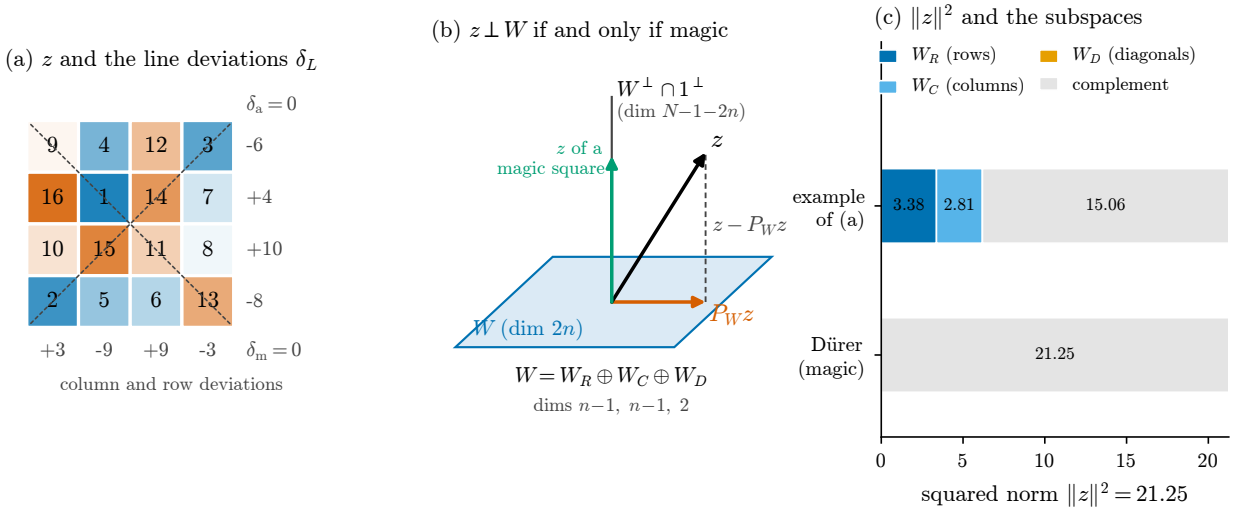


Figure 1: The line space for $n = 4$. W is the $2n$-dimensional span of the centred line indicators, and $W^\perp \cap \mathbf{1}^\perp$ (dimension $N - 1 - 2n$) is its orthogonal complement among the zero-mean functions. (a) The arrangement of Remark 2.5, shaded by z, with its row and column deviations (both diagonal deviations are 0). (b) Schematic of the orthogonal splitting $z = P_W z + (z - P_W z)$; the arrangement is magic if and only if $P_W z = 0$. (c) The squared norms $\|P_{W_R} z\|^2, \|P_{W_C} z\|^2, \|P_{W_D} z\|^2$ and that of the complement, for the arrangement of (a) $(3.38 + 2.81 + 0 + 15.06)$ and for Dürer's magic square (all of $\|z\|^2 = 21.25$ lies in the complement).

centred array, and assume $d_N \neq 0$. By [15, Thm. 3], $\sum_i c_N(i, \pi(i))$ is asymptotically normal, i.e. $\big(\sum_i c_N(i, \pi(i)) - \mathbb{E}\big)/\sqrt{\mathrm{Var}}$ converges in distribution to $N(0, 1)$, if

$$\lim_{N \to \infty} \frac{N^{-1} \sum_{i,j} d_N(i, j)^r}{\big(N^{-1} \sum_{i,j} d_N(i, j)^2\big)^{r/2}} = 0, \qquad r = 3, 4, \ldots, \tag{8}$$

and (8) is satisfied if

$$\lim_{N \to \infty} \frac{\max_{i,j} d_N(i, j)^2}{N^{-1} \sum_{i,j} d_N(i, j)^2} = 0 \tag{9}$$

(conditions (11) and (12) of [15]). For a product array $c_N(i, j) = a_i b_j$ one has $d_N(i, j) = (a_i - \bar{a})(b_j - \bar{b})$, and (9) becomes

$$\lim_{N \to \infty} N \, \frac{\max_i (a_i - \bar{a})^2 \, \max_j (b_j - \bar{b})^2}{\sum_i (a_i - \bar{a})^2 \, \sum_j (b_j - \bar{b})^2} = 0,$$

the sufficient condition (14) of Hoeffding's Theorem 4, which treats this case. Motoo [24] gives a Lindeberg-type condition and Bolthausen [6] a Berry–Esseen bound.

2.4 Known counts

There are 8 magic squares of order 3, all images of the Lo Shu under D_4; 7,040 of order 4, forming the 880 D_4-orbits listed by Frénicle de Bessy [13] and recounted many times since [4]; and 275,305,224 D_4-orbits of order 5, i.e. 2,202,441,792 squares, counted by Schroeppel in 1973 and reported by Gardner [14]. For $n \geq 6$ only statistical estimates are known [29, 35].

3 The moment hierarchy and the permutation threshold

3.1 Moment conditions

For a centred $f \in \mathcal{F}$ let $\gamma_j(f) = \sum_i f_{ij}$ and $\rho_i(f) = \sum_j f_{ij}$ be its column and row sums; for $f = z(S)$ these are the deviations γ, ρ of S. For $\gamma \in \mathbb{R}^n$ put $P_\gamma(t) = \sum_{j=0}^{n-1} \gamma_j t^j$.

Definition 3.1 (Degree-k conditions)**.** A centred $f \in \mathcal{F}$ satisfies the *degree-k conditions* ($k \geq 1$) if $\langle x^m, f\rangle = \langle y^m, f\rangle = 0$ for $1 \leq m \leq k$. For $f = z(S)$ these read $\sum_{i,j} z_{ij} x_j^m = \sum_{i,j} z_{ij} y_i^m = 0$, i.e. $\mathrm{Cov}(X^m, Z) = \mathrm{Cov}(Y^m, Z) = 0$.

Lemma 3.2 (Polynomial form)**.** *If φ depends only on the column index, $\langle \varphi, f\rangle = \frac{1}{N}\sum_j \varphi(j)\gamma_j(f)$; similarly for rows. For $\gamma \in \mathbb{R}^n$ and $k \geq 0$ the following are equivalent: (i) $\sum_j x_j^m \gamma_j = 0$ for $0 \leq m \leq k$; (ii) $\sum_j \binom{j}{r}\gamma_j = 0$ for $0 \leq r \leq k$; (iii) $(t-1)^{k+1}$ divides $P_\gamma(t)$. Hence a centred f satisfies the degree-k conditions if and only if $(t-1)^{k+1}$ divides both $P_{\gamma(f)}$ and $P_{\rho(f)}$.*

Proof. The first statement is $\sum_{i,j}\varphi(j) f_{ij} = \sum_j \varphi(j)\gamma_j(f)$. The families $(x_j^m)_{m\leq k}$ and $\bigl(\binom{j}{r}\bigr)_{r\leq k}$ both span the restrictions to $\{0,\dots,n-1\}$ of the polynomials of degree at most k in j, so (i)$\Leftrightarrow$(ii); and $P_\gamma(t) = \sum_r \bigl(\sum_j \binom{j}{r}\gamma_j\bigr)(t-1)^r$ is the Taylor expansion at $t=1$, so (ii)$\Leftrightarrow$(iii). The case $m = 0$ of (i) holds for every centred f. □

Proposition 3.3 (Moment hierarchy)**.** *Let $n \geq 3$ and let $f \in \mathcal{F}$ be centred.*

(a) If $\gamma(f) = \rho(f) = 0$, then $\langle a(X) + b(Y), f\rangle = 0$ for all functions a, b on $\{x_0, \dots, x_{n-1}\}$; in particular f satisfies the degree-k conditions for every k. For $f = z(S)$ with S magic, also $\langle a(X) + b(Y) + c_1 \mathbf{1}_{D_\mathrm{m}} + c_2 \mathbf{1}_{D_\mathrm{a}}, z\rangle = 0$.

(b) If $k \geq n-1$, the degree-k conditions imply $\gamma(f) = \rho(f) = 0$.

(c) If $1 \leq k \leq n-2$, the function $f_{ij} = \frac{1}{n}(-1)^j\binom{k+1}{j}$ ($0 \leq j \leq k+1$), $f_{ij} = 0$ ($j > k+1$), satisfies the degree-k conditions but not the degree-$(k+1)$ conditions, and $\gamma(f) \neq 0$.

Proof. (a) is the first statement of Lemma 3.2. (b) A nonzero polynomial of degree at most $n-1$ is not divisible by $(t-1)^n$. (c) Here $\rho(f) = 0$ and $P_{\gamma(f)}(t) = (1-t)^{k+1}$. □

Part (a) contains the vanishing of the centre of mass $\sum_c z_c(x_c, y_c)$ and of all axial moments $\sum_c z_c x_c^m$, $\sum_c z_c y_c^m$ of a semi-magic square, in the spirit of the multipole analysis of a magic cube by Rogers and Loly [31]. Parts (b) and (c) are the invertibility of a Vandermonde matrix. The mixed moments are not determined by the line sums:

Remark 3.4 (Mixed moments). The magic square with rows $(1,6,11,16)$, $(8,15,2,9)$, $(12,3,14,5)$, $(13,10,7,4)$ has $\sum_c z_c x_c y_c = 60$. Among the 880 magic squares of order 4 up to D_4, the product of inertia $\sum_c z_c x_c y_c$ is nonzero for 824, takes 48 distinct absolute values between 0 and 60, and $\sum_c z_c x_c^2 y_c$, $\sum_c z_c x_c y_c^2$ are nonzero for 623 and 773 of them. For $n = 3$ the mixed moment $\sum_c z_c x_c y_c$ vanishes for all eight magic squares, because $s_c + s_{-c} = 10$ for centrally opposite cells, so z is odd under $(x, y) \mapsto (-x, -y)$ while xy is even.

Corollary 3.5 (Loly's invariant [19])**.** *Let $I(S) = \sum_{i,j} s_{ij}(x_j^2 + y_i^2)$ be the moment of inertia about the centre of the masses s_{ij} placed at (x_j, y_i). For every arrangement,*

$$I(S) = N\,\mathrm{Cov}\bigl(Z,\, X^2 + Y^2\bigr) + \frac{n^2(n^4-1)}{12} = \sum_j x_j^2\gamma_j + \sum_i y_i^2\rho_i + \frac{n^2(n^4-1)}{12}. \tag{10}$$

Hence $I(S) = n^2(n^4-1)/12$ if and only if the single condition $\mathrm{Cov}(Z, X^2+Y^2) = 0$ holds; it is implied by the degree-2 conditions, hence by the semi-magic property.

Proof. Substitute $s_{ij} = z_{ij} + \frac{N+1}{2}$ and use Lemma 2.2 and Lemma 3.2: $\frac{N+1}{2}\sum_{i,j}(x_j^2 + y_i^2) = \frac{N+1}{2}\cdot\frac{n^2(n^2-1)}{6}$. □

Loly's derivation uses the row and column sums in the same way [19]. The single condition is much weaker than the degree-2 conditions: for $n = 3$, 24,192 of the 9! arrangements have $I = 60$, while 72 (the semi-magic squares, Remark 3.15) satisfy the degree-2 conditions. The energy E_{low} of (4) combines the degree-1 conditions with the two diagonal conditions: $E_{\mathrm{low}} = \langle x, z\rangle^2 + \langle y, z\rangle^2 + (\delta_\mathrm{m}^2 + \delta_\mathrm{a}^2)/N^2$.

3.2 The permutation threshold

By Proposition 3.3, for real-valued f the degree-k conditions force equal row and column sums exactly when $k \geq n-1$. For arrangements the deviations are integers of bounded size, and the threshold drops.

Definition 3.6. For $n \geq 3$, $k^*(n)$ is the least $k \geq 1$ such that every arrangement of order n satisfying the degree-k conditions is semi-magic.

By Proposition 3.3(b), $k^*(n) \leq n-1$. By Lemma 3.2, an arrangement satisfies the degree-k conditions if and only if γ and ρ lie in the lattice

$$\Lambda_{n,k} = \{\gamma \in \mathbb{Z}^n : (t-1)^{k+1} \mid P_\gamma(t)\} = (t-1)^{k+1}\,\mathbb{Z}[t]_{<n-k-1},$$

of rank $n-k-1$. Writing $\gamma = \gamma^+ - \gamma^-$ with $\gamma^\pm \geq 0$, the condition $\gamma \in \Lambda_{n,k}$ says that the multisets in which j occurs γ_j^+, resp. γ_j^-, times have equal power sums of degrees $0, \ldots, k$: a Prouhet–Tarry–Escott system [38, 7]. Small vectors in $\Lambda_{n,k}$ are classical: the Prouhet–Thue–Morse vector $\big((-1)^{s_2(j)}\big)_{0\leq j<2^{k+1}}$ (s_2 the binary digit sum) has entries ± 1 and lies in $\Lambda_{2^{k+1},k}$ [38, 2], and Borwein, Erdélyi and Kós construct polynomials of degree n with coefficients in $\{-1,0,1\}$ vanishing to order $\geq c\sqrt{n/\log n}$ at 1 [8, Thm. 2.7]. The question is therefore how large k can be for vectors that are column-deviation vectors of arrangements.

Theorem 3.7 (The permutation threshold)**.**

(a) For all $n \geq 3$, $k^*(n) \leq \left\lfloor \sqrt{\pi e\,(n-1)\log(n^4/4)} \right\rfloor \leq 2\sqrt{\pi e}\,\sqrt{n\log n}$.

(b) $\displaystyle\liminf_{n\to\infty} \frac{k^*(n)}{\sqrt{n}} \geq \sqrt{2}$.

(c) $k^*(n) = n-1$ for $3 \leq n \leq 13$, $k^*(n) = n-2$ for $14 \leq n \leq 17$, and $k^*(n) \leq n-3$ for every $n \geq 18$. The values for $14 \leq n \leq 40$ are those of Table 1.

The constant $2\sqrt{\pi e}$ is about 5.84. Part (a) improves on $k^*(n) \leq n-3$ only for $n \geq 166$; for smaller n the exact computation behind (c) is much stronger (Figure 2). Parts (c) for $n \geq 14$ and the values in Table 1 rest on exact computer calculations, described in Section 3.5.

3.3 Proof of the upper bound

Lemma 3.8 (Column feasibility)**.** *Let γ be the column-deviation vector of an arrangement. For every set J of p columns,*

$$\Big|\sum_{j\in J} \gamma_j\Big| \leq \frac{pn(N-pn)}{2}. \tag{11}$$

In particular $|\gamma_j| \leq n^2(n-1)/2$, $\|\gamma\|_1 \leq N^2/4$ and $\|\gamma\|_2^2 \leq n^5(n^2-1)/12$. The same holds for ρ.

Proof. The pn cells of J carry distinct values, whose sum is at most that of the pn largest values, $pn(2N-pn+1)/2$; subtracting $pM(n) = pn(N+1)/2$ gives the upper bound, and the lower bound is symmetric. With p the number of positive γ_j, $\|\gamma\|_1 = 2\sum_{\gamma_j>0}\gamma_j \leq pn(N-pn) \leq N^2/4$. Finally (11) says that γ is majorised by the vector with entries $n(N-(2p-1)n)/2$, $1 \leq p \leq n$, whose squared norm is $n^5(n^2-1)/12$, and $\|\cdot\|_2^2$ is Schur-convex. □

Lemma 3.9 (Order of vanishing of an integer polynomial at 1)**.** *Let $\gamma \in \mathbb{Z}^n \setminus \{0\}$ with $(t-1)^K \mid P_\gamma$ for some $K \geq 1$. Then $K^2 \leq \pi e\,(n-1)\log\|\gamma\|_1$.*

Proof. Write $P = P_\gamma$ and $L = \|\gamma\|_1$; note $K \le \deg P \le n-1$. Since P has integer coefficients and leading coefficient $a \neq 0$, Jensen's formula gives $\int_{-1/2}^{1/2} \log|P(e^{2\pi i\theta})|\,d\theta = \log\bigl(|a|\prod_\alpha \max(1,|\alpha|)\bigr) \ge 0$, the product over the roots of P. On $|z|=1$ we have $|P(z)| \le L$. Since $P^{(i)}(1) = 0$ for $i < K$, Taylor's formula with integral remainder along the segment $[1,z]$, which lies in the closed unit disc when $|z|=1$, gives

$$|P(z)| \le \frac{|z-1|^K}{K!}\max_{|w|\le 1}|P^{(K)}(w)| \le L\binom{n-1}{K}|z-1|^K \le L\Bigl(\frac{e(n-1)|z-1|}{K}\Bigr)^K,$$

because $|P^{(K)}(w)| \le \sum_j |\gamma_j|\,j(j-1)\cdots(j-K+1)$ for $|w| \le 1$. With $|e^{2\pi i\theta}-1| \le 2\pi|\theta|$ and $\beta = 2\pi e(n-1)/K \ge 2\pi e$ we get $\log|P(e^{2\pi i\theta})| \le \log L + K\log\min(1,\beta|\theta|)$, and $\int_{-1/2}^{1/2}\log\min(1,\beta|\theta|)\,d\theta = -2/\beta$. Hence $0 \le \log L - 2K/\beta = \log L - K^2/(\pi e(n-1))$. □

Proof of Theorem 3.7(a). Let S satisfy the degree-k conditions with $(k+1)^2 > \pi e(n-1)\log(n^4/4)$. If $\gamma \neq 0$, then $(t-1)^{k+1} \mid P_\gamma$ (Lemma 3.2) and $\|\gamma\|_1 \le n^4/4$ (Lemma 3.8), contradicting Lemma 3.9. So $\gamma = 0$, and likewise $\rho = 0$. The second inequality follows from $(n-1)\log(n^4/4) < 4n\log n$. □

The order $\sqrt{n\log n}$ also follows from [8, Thm. 2.1], applied to $P_\gamma(t)/(t^r\max_j|\gamma_j|)$ with r the index of the first nonzero γ_j; that theorem bounds the number of zeros at 1 of $\sum_{j\le n} a_j t^j$ with $|a_j| \le 1$ by $c\sqrt{n(1-\log|a_0|)}$, with an unspecified absolute constant. Lemma 3.9 is a self-contained version for integer coefficients. An argument using only that γ is a nonzero integer vector with $\|\gamma\|_\infty \le n^3$ cannot give an upper bound of smaller order than $\sqrt{n}$: the box principle applied to the vectors in $\{0,\dots,n^3\}^n$ and their binomial moments of order $< K$ produces such vectors in $\Lambda_{n,K-1}$ with $K \asymp \sqrt{n}$.

3.4 Proof of the lower bound

A lower bound $k^*(n) \ge K$ requires an arrangement that satisfies the degree-$(K-1)$ conditions without being semi-magic. We keep all row sums equal to $M(n)$ and control the column deviations by swaps.

Definition 3.10. Rows $i \neq i'$ of an arrangement are *twins on* a set J of columns if $s_{ij}+s_{i'j} = N+1$ for all $j \in J$.

Lemma 3.11 (Twin swaps)**.** *Let S be semi-magic, let rows i, i' be twins on J, let $j, j' \in J$ and $d = s_{ij'} - s_{ij}$. Exchanging the entries of row i in columns j, j' changes the column deviations by $d(e_j - e_{j'})$, exchanging those of row i' changes them by $-d(e_j - e_{j'})$, and neither changes a row sum. Exchanges involving pairwise disjoint sets of cells can be combined, and their effects add.*

Proof. $s_{i'j} - s_{i'j'} = (N+1-s_{ij}) - (N+1-s_{ij'}) = d$. □

Lemma 3.12 (Squares with many twin rows)**.**

1. *If $n = 2m$ with m even, or with m odd and $m \ge 7$, there is a semi-magic square of order n in which rows r and $m+r$ are twins on all columns, for $0 \le r < m$.*
2. *If $n \ge 9$ is odd, there is a semi-magic square of order n in which rows ℓ and $n-1-\ell$ are twins on the columns $\ell+1,\dots,n-2-\ell$, for $0 \le \ell \le (n-9)/2$.*

The proof is a direct construction, given in Section B.

Proposition 3.13 (Box-principle criterion)**.** *Let S be a square of Lemma 3.12. Pair the columns as (j_q, j_q+1), where $j_q = 2q$ for $q < Q = n/2$ (n even) and $j_q = 2q+1$ for $q < Q = (n-3)/2$*

(n odd), and let R_q be the number of twin pairs of rows that are twins on both columns of the q-th pair: $R_q = n/2$ for even n, and $R_q = \min(2q,\, n-4-2q,\, (n-9)/2)+1$ for odd n. If $K \ge 2$ and

$$\prod_{q<Q}\Big(\Big\lceil\frac{R_q}{2}\Big\rceil+1\Big) \;>\; \prod_{r=0}^{K-2}\Big((N-1)\sum_{q<Q}R_q\binom{q}{r}+1\Big), \tag{12}$$

then there is an arrangement with all row sums $M(n)$ and $\gamma \neq 0$ that satisfies the degree-$(K-1)$ conditions; hence $k^*(n) \ge K$.

Proof. For each q, Lemma 3.11 gives R_q nonzero integers $d_{q,1},\dots,d_{q,R_q}$ with $|d_{q,i}| \le N-1$, one for each twin pair covering the q-th column pair. At least $\lceil R_q/2\rceil$ of them have the same sign, and their partial sums, together with 0, form a set F_q of $\lceil R_q/2\rceil+1$ distinct subset sums lying in an interval of length at most $R_q(N-1)$. For $g \in \prod_q F_q$ the integers $\mu_r(g) = \sum_q g_q\binom{q}{r}$, $0 \le r \le K-2$, lie in intervals of lengths at most $(N-1)\sum_q R_q\binom{q}{r}$, so by (12) there are $g \neq g'$ with $\mu_r(g) = \mu_r(g')$ for all $r \le K-2$. Write $g_q = \sum_{i\in I_q} d_{q,i}$ and $g'_q = \sum_{i\in I'_q} d_{q,i}$. Perform the row-$i$ exchange of Lemma 3.11 for the twin pairs in $I_q \setminus I'_q$ and the twin-row exchange for those in $I'_q \setminus I_q$, for every q; all these exchanges involve disjoint cells. The result has all row sums $M(n)$ and $\gamma = \sum_q h_q(e_{j_q} - e_{j_q+1})$ with $h = g - g' \neq 0$. Then $P_\gamma(t) = t^{j_0}(1-t)H(t^2)$ with $H(u) = \sum_q h_q u^q \neq 0$. Since $\sum_q h_q\binom{q}{r} = 0$ for $r \le K-2$, Lemma 3.2 gives $(u-1)^{K-1} \mid H(u)$, so $(t^2-1)^{K-1} \mid H(t^2)$ and $(t-1)^K \mid P_\gamma$. With $\rho = 0$, the arrangement satisfies the degree-$(K-1)$ conditions, and $\gamma \neq 0$. □

Proof of Theorem 3.7(b). Fix $\alpha < \sqrt{2}$ and let $K-1 = k = \lfloor\alpha\sqrt{n}\rfloor$; we show that (12) holds for all large n covered by Lemma 3.12 (every $n \ge 14$ is covered). *Right side.* $\sum_{q<Q}\binom{q}{r} = \binom{Q}{r+1} \le \binom{n}{r+1}$, $R_q \le n$ and $N \le n^2$, so each factor is at most $n^4\binom{n}{r+1}$, and with $\log s! \ge s\log s - s$,

$$\log(\text{right side}) \le 4k\log n + \sum_{s=1}^{k}\big(s\log n - \log s!\big) \le \frac{k^2}{2}\log\frac{n}{k} + O(k^2 + k\log n) = \frac{\alpha^2}{4}\,n\log n + O(n).$$

Left side. For even n it equals $(\lceil n/4\rceil+1)^{n/2}$, with logarithm $\frac{n}{2}\log n - O(n)$. For odd n, $R_q \ge \min(q+1,\, Q-q,\, (n-9)/2)$, and $\min(q+1, Q-q) \le (Q+1)/2 < (n-9)/2$ for large n, so the logarithm is at least $2\log(\lfloor Q/2\rfloor!) - O(n) = \frac{n}{2}\log n - O(n)$. Since $\alpha^2/4 < 1/2$, (12) holds for large n, and $k^*(n) \ge \lfloor\alpha\sqrt{n}\rfloor + 1$. □

The convergence in (b) is slow: (12) itself gives only $k^*(n) \ge 2$ at $n = 14$ and $k^*(n) \ge 7$ at $n = 100$. Thus

$$(\sqrt{2} - o(1))\sqrt{n} \le k^*(n) \le (2\sqrt{\pi e} + o(1))\sqrt{n\log n},$$

and the factor $\sqrt{\log n}$ between the bounds is the analogue of the gap between $\sqrt{n/\log n}$ and $\sqrt{n}$ left open in [8] for polynomials with coefficients in $\{-1, 0, 1\}$.

3.5 Exact values

Proof of Theorem 3.7(c). $3 \le n \le 13$. Here $k^*(n) \le n-1$, and the lower bound needs an arrangement satisfying the degree-$(n-2)$ conditions that is not semi-magic. For $n = 3$ and $n = 4$,

$$\begin{pmatrix}1&3&9\\8&6&5\\7&4&2\end{pmatrix}\quad \gamma = (1,-2,1),\ \rho = (-2,4,-2); \qquad \begin{pmatrix}14&2&5&13\\8&15&7&4\\12&3&9&10\\1&11&16&6\end{pmatrix}\quad \gamma = (1,-3,3,-1),\ \rho = 0.$$

In both cases $P_\gamma = (1-t)^{n-1}$ and $P_\rho \in \{0, -2(1-t)^2\}$, so the degree-$(n-2)$ conditions hold. For $5 \le n \le 13$ we found arrangements with $\rho = 0$ and $\gamma_j = (-1)^j\binom{n-1}{j}$, i.e. $P_\gamma = (1-t)^{n-1}$ (Section D).

$n \geq 14$: *level* $n-2$. $\Lambda_{n,n-2}$ has rank 1 and is spanned by $v = \left((-1)^j \binom{n-1}{j}\right)_j$. If $\gamma = tv \neq 0$ with $t \in \mathbb{Z}$, then $\max_j |\gamma_j| \geq \binom{n-1}{\lfloor (n-1)/2 \rfloor}$, which exceeds $n^2(n-1)/2$ for $n \geq 14$, contradicting Lemma 3.8. (At $n = 14$, $\binom{13}{6} = 1716 > 1274$; passing from n to $n+1$ multiplies the left side by 2 for even n and by $2n/(n+1) \geq 15/8$ for odd $n \geq 15$, and the right side by $(n+1)^2/(n(n-1))$, which decreases in n and equals $225/182 < 15/8$ at $n = 14$.) So $k^*(n) \leq n-2$ for $n \geq 14$; at $n = 13$ the bound fails ($\binom{12}{6} = 924 < 1014$), and the witness has central entry 924.

$n \geq 18$: *level* $n-3$. $\Lambda_{n,n-3}$ has rank 2: $\gamma = (t-1)^{n-2}(a+bt)$ with $a, b \in \mathbb{Z}$. Two consecutive central coefficients bound $|a|, |b|$, and checking all pairs (a, b) in the resulting box shows that for $18 \leq n \leq 400$ no nonzero $\gamma \in \Lambda_{n,n-3}$ satisfies $|\gamma_j| \leq n^2(n-1)/2$. For $n \geq 166$, part (a) gives $k^*(n) \leq n-3$, since $f(n) = (n-2)^2 - \pi e(n-1)\log(n^4/4)$ is positive at $n = 166$ and $f'(n) \geq 2(n-2) - 4\pi e(\log n + 1) > 0$ for $n \geq 166$ (the last expression increases for $n \geq 18$ and is positive at $n = 166$).

Upper bounds for $14 \leq n \leq 40$. If no nonzero vector of $\Lambda_{n,k}$ satisfies (11) for every p, then $k^*(n) \leq k$. By Lemma 3.8 it suffices to examine the vectors of $\Lambda_{n,k}$ with $\|\gamma\|_2^2 \leq n^5(n^2-1)/12$; we enumerated them exactly (LLL reduction followed by Fincke–Pohst enumeration in rational arithmetic, Section A). At the level $k = k^*(n)$ of Table 1, no nonzero vector passes the test.

Lower bounds for $14 \leq n \leq 40$. For each n we found an arrangement with all row sums $M(n)$ whose column deviations satisfy $(t-1)^{k^*(n)} \mid P_\gamma$ and $\gamma \neq 0$; each was verified in exact integer arithmetic. □

n	14	15	16	17	18	19	20	21	22	23	24	25	26	27
$k^*(n)$	12	13	14	15	15	16	16	17	17	18	18	19	19	20
$\max_j \lvert\gamma_j\rvert$	297	572	1001	2002	1430	2860	1794	3094	2210	3638	2380	4522	2736	4541
$n^2(n-1)/2$	1274	1575	1920	2312	2754	3249	3800	4410	5082	5819	6624	7500	8450	9477

n	28	29	30	31	32	33	34	35	36	37	38	39	40
$k^*(n)$	21	21	22	22	23	23	23	24	24	25	25	26	26
$\max_j \lvert\gamma_j\rvert$	8892	6405	11737	9330	14881	11799	9177	18147	14766	24150	16515	26520	22233
$n^2(n-1)/2$	10584	11774	13050	14415	15872	17424	19074	20825	22680	24642	26714	28899	31200

Table 1: Exact values of the permutation threshold for $14 \leq n \leq 40$ (Theorem 3.7(c)), the largest column deviation $\max_j |\gamma_j|$ of the witness arrangement, and the bound $n^2(n-1)/2$ of Lemma 3.8. The upper bounds come from the exact lattice enumeration of `anc/kstar_verify.py`; for $n \leq 33$ they were also reproduced by an independent integer-programming computation (`anc/kstar_crosscheck/`), so for $34 \leq n \leq 40$ they rest on the lattice enumeration alone.

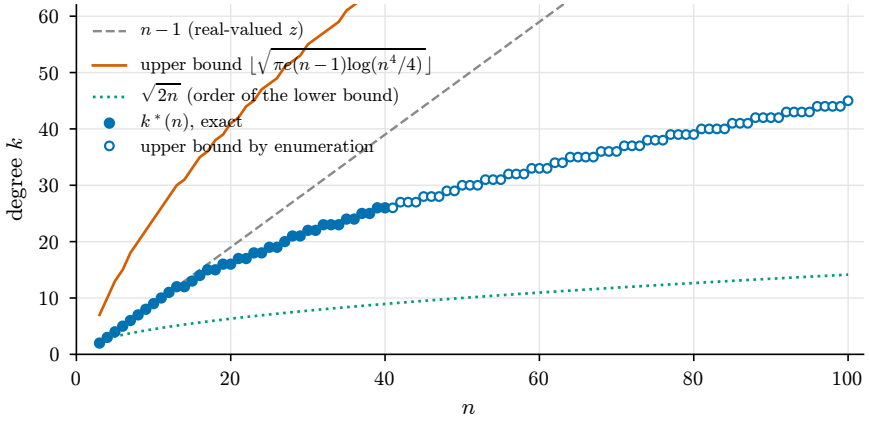


Figure 2: The permutation threshold. Exact values of $k^*(n)$ for $3 \leq n \leq 40$ (dots), the upper bounds obtained by the exact lattice enumeration for $41 \leq n \leq 100$ (open circles), the linear-algebra threshold $n-1$, the bound of Theorem 3.7(a), and the curve $\sqrt{2n}$, the asymptotic order of the lower bound (b).

The witnesses. The target vectors γ were chosen among the shortest vectors of $\Lambda_{n,k^*(n)-1}$ satisfying (11), as produced by the enumeration. Starting from a semi-magic square of Lemma 3.12

(or a Siamese square), a steepest-descent search over exchanges within a row, which fix the row sums, minimised $\sum_j (\text{column sum}_j - M(n) - \gamma_j)^2$, with random exchanges to leave local minima; for the hardest cases a simulated annealing over arbitrary exchanges, with energy the squared distance of both row and column sums to their targets, was used. We observe (for these n only; this is not a theorem) that for $14 \le n \le 40$ the threshold is determined by the column-feasibility inequalities alone: $k^*(n)$ is the least k for which no nonzero vector of $\Lambda_{n,k}$ satisfies (11). Some witnesses come close to the bound of Lemma 3.8: $\max_j |\gamma_j| = 924$ against 1014 at $n = 13$, and 24150 against 24642 at $n = 37$. Beyond $n = 40$ we computed only upper bounds (Figure 2); for instance $k^*(50) \le 30$, $k^*(80) \le 39$ and $k^*(100) \le 45$.

3.6 The low-order conditions for $n = 3$

Proposition 3.14. *For $n = 3$, $E_{\text{low}}(S) = 0$ if and only if S is magic.*

Proof. Write the centred entries $z_c = s_c - 5 \in \{-4, \dots, 4\}$ as $\left(\begin{smallmatrix} a & b & c \\ d & e & f \\ g & h & i \end{smallmatrix}\right)$, so that $\sum_c z_c = 0$, $\sum_c z_c^2 = 60$, and every line sum of z is the deviation of that line. The degree-1 conditions with $\sum_j \gamma_j = \sum_i \rho_i = 0$ give $\gamma = (t, -2t, t)$ and $\rho = (u, -2u, u)$, and the diagonal conditions give $a + e + i = c + e + g = 0$. The middle row, the middle column and the two diagonals cover every cell once and the centre three more times, so $3e = -2(t + u)$; as e is an integer, $t + u = 3k$ and $e = -2k$ with $k \in \mathbb{Z}$. Put $P = a - k$, $Q = c - k$, $T = t - k$, $U = u - k$, so that $T + U = k$. The diagonals give $i = k - P$, $g = k - Q$; the first and last rows (sum u) and columns (sum t) then give

$$z = \begin{pmatrix} k+P & -T-P-Q & k+Q \\ -U-P+Q & -2k & -U+P-Q \\ k-Q & -T+P+Q & k-P \end{pmatrix}, \qquad \sum_c z_c^2 = 2\big(4k^2 + 3(P^2+Q^2) + T^2 + U^2\big),$$

so $4k^2 + 3(P^2 + Q^2) + T^2 + U^2 = 30$. The four corners are distinct, so $P \ne 0$, $Q \ne 0$ and $|P| \ne |Q|$, hence $P^2 + Q^2 \ge 5$; therefore $R = 4k^2 + T^2 + U^2$ is a multiple of 3 and at most 15, and $P^2 + Q^2 = 10 - R/3$. So $|k| \le 1$. If $|k| = 1$, then $T + U = \pm 1$ makes $T^2 + U^2$ odd, so $R \in \{9, 15\}$ (as $R \ge 5$) and $T^2 + U^2 \in \{5, 11\}$; 11 is not a sum of two squares, and $T^2 + U^2 = 5$ gives $P^2 + Q^2 = 7$, which is not either. So $k = 0$, $U = -T$, and $R = 2T^2 \le 15$ divisible by 3 forces $T = 0$. Hence $t = u = 0$: all rows, columns and diagonals have deviation 0, and S is magic. The converse is clear. □

Remark 3.15. The enumeration of all 9! arrangements confirms Proposition 3.14 and each step of its proof (`anc/n3_lowmode_proof.py`): exactly 760 arrangements satisfy the degree-1 conditions, 4,224 have both diagonal sums 15, 72 (the semi-magic squares) satisfy the degree-2 conditions, and 8, the magic squares, satisfy the degree-1 and both diagonal conditions; the only integer solution of $4k^2 + 3(P^2 + Q^2) + T^2 + U^2 = 30$ with $T + U = k$, $PQ \ne 0$, $|P| \ne |Q|$ is $k = T = U = 0$, $P^2 + Q^2 = 10$. The four conditions are weaker than the magic conditions for real-valued z; the proof uses that the entries are distinct integers with $\sum_c z_c^2 = 60$. For $n = 4$ the equivalence fails (Remark 2.5); the arrangement there has $\gamma = (3, -9, 9, -3)$ and $\rho = (-6, 4, 10, -8)$, so it satisfies the degree-2 condition in x but not in y.

3.7 Balance of the signed mass

Regard z_c as a signed mass at (x_c, y_c). By Proposition 3.3(a), a semi-magic square balances every separable test function $a(x) + b(y)$, and a magic square also the diagonal indicators; by Remark 3.4 it need not balance xy. For arbitrary arrangements the balance is controlled by the energy:

Remark 3.16 (Quantitative balance). Since $\sum_c z_c x_c = \sum_j x_j \gamma_j$ and $\sum_j x_j^2 = n(n^2 - 1)/12$, the Cauchy–Schwarz inequality gives

$$\Big|\sum_c z_c (x_c, y_c)\Big|^2 \le \frac{n(n^2-1)}{12}\big(\|\gamma\|^2 + \|\rho\|^2\big) \le N^2 \, \frac{n(n^2-1)}{12}\, E_{\text{full}},$$

with equality in the first step if and only if $\gamma \parallel (x_j)$ and $\rho \parallel (y_i)$, for instance for $s_{ij} = in + j + 1$. More generally, by Lemma 2.3, $\|P_W z\|^2 = n(\|\rho\|^2 + \|\gamma\|^2)/N^2 + \frac{1}{N}\delta^{\mathsf{T}}\Gamma_D^{-1}\delta$ with $\delta = (\delta_{\mathrm{m}}, \delta_{\mathrm{a}})$, so $nE_{\mathrm{full}} \le \|P_W z\|^2 \le \lambda_n E_{\mathrm{full}}$ with $\lambda_n = n^2/(n-1)$ for odd n and $n^2/(n-2)$ for even n; hence $|\sum_c z_c\varphi_c| \le N\|\varphi\|\sqrt{\lambda_n E_{\mathrm{full}}}$ for every $\varphi \in W$.

Nonlinear balance is not implied. With the radial unit vectors $u_c = (x_c, y_c)/|(x_c, y_c)|$ ($u_c = 0$ at the centre), each of the eight 3×3 magic squares has $|\sum_c z_c u_c| = 2(2-\sqrt{2})\sqrt{5} \approx 2.62$: for the Lo Shu with rows $(2,7,6), (9,5,1), (4,3,8)$ the edge cells contribute $(-8, 4)$ and the corners $(4\sqrt{2}, -2\sqrt{2})$. For $n = 4$ this ring imbalance is positive for all 880 magic squares (smallest value 0.070), with mean 5.51 against 17.06 for uniformly random arrangements; the magic mean lies at the 7.4th percentile of the random distribution (`anc/balance_check.py`).

4 The energy of a random arrangement

Let S be a uniformly random arrangement: $z_c = b_{\pi(c)}$ with π uniform and b the centred values $k - \frac{N+1}{2}$, $1 \le k \le N$. For a test function φ put $T_\varphi = \mathrm{Cov}(\varphi, Z) = \sum_c a_c z_c$ with $a_c = (\varphi_c - \bar{\varphi})/N$, a linear permutation statistic (Section 2.3). $\mathrm{Var}(\varphi)$ and $\mathrm{Cov}(\varphi, \psi)$ denote population variances and covariances over the N cells, and $\sigma_n^2 = (n^2+1)/12$ as in (7).

4.1 Means

Proposition 4.1 (Means and covariance structure)**.** *For all test functions φ, ψ, $\mathbb{E}[T_\varphi] = 0$ and $\mathbb{E}[T_\varphi T_\psi] = \sigma_n^2 \mathrm{Cov}(\varphi, \psi)$. Consequently*

$$\mathbb{E}\,\mathrm{Cov}(X, Z)^2 = \mathbb{E}\,\mathrm{Cov}(Y, Z)^2 = \frac{n^4 - 1}{144}, \qquad \mathbb{E}\,\mathrm{Cov}(\mathbf{1}_L, Z)^2 = \frac{(n-1)(n^2+1)}{12\,n^2} \quad \text{for every line } L, \tag{13}$$

$$\mathbb{E}[E_{\mathrm{low}}] = \frac{(n-1)(n^2+1)(n^3+n^2+12)}{72\,n^2}, \qquad \mathbb{E}[E_{\mathrm{full}}] = \frac{n^4-1}{6\,n^2}. \tag{14}$$

The covariance matrix of the vector of statistics in E_{low} (resp. E_{full}) is $\sigma_n^2\Gamma$, with Γ the Gram matrix of the centred test functions; its eigenvalues are

$$E_{\mathrm{low}}:\ \sigma_n^2\Big\{\tfrac{n^2-1}{12}\,(\times 2),\ \tfrac{n-1}{n^2}\,(\times 2)\Big\}, \qquad E_{\mathrm{full}}:\ \sigma_n^2\Big\{\tfrac{1}{n}\,(\times 2n-2),\ \tfrac{n-1}{n^2}\,(\times 2),\ 0\,(\times 2)\Big\} \tag{15}$$

for odd n; for even n the pair $\{\frac{n-1}{n^2}, \frac{n-1}{n^2}\}$ is replaced by $\{\frac{1}{n}, \frac{n-2}{n^2}\}$.

Proof. $\sum_c a_c = 0$ gives $\mathbb{E}T_\varphi = 0$, and (6) with $\sum_j b_j^2 = N\mathrm{Var}(Z)$ gives $\mathbb{E}[T_\varphi T_\psi] = \frac{1}{N-1} \cdot \frac{\mathrm{Cov}(\varphi,\psi)}{N} \cdot N\mathrm{Var}(Z) = \sigma_n^2 \mathrm{Cov}(\varphi, \psi)$. The entries of Γ follow from Lemma 2.3: $\mathrm{Var}(\mathbf{1}_L) = (n-1)/n^2$, two lines meeting in ℓ cells have covariance $\ell/N - 1/n^2$, and X, Y are uncorrelated with each other and with the diagonal indicators, since a diagonal meets every row and every column once. The rows give the block $\frac{1}{n}(I - \frac{1}{n}J)$, the columns likewise, and the means are the traces. □

Equivalently, a line sum is the sum of a sample of size n drawn without replacement from $\{1, \dots, N\}$, so $\mathbb{E}\delta_L^2 = n\mathrm{Var}(Z)\frac{N-n}{N-1}$. The means of the two energies coincide if and only if $\frac{n^2-1}{6} = \frac{2(n-1)}{n}$, i.e. $n = 3$, where both equal $40/27$.

Corollary 4.2 (Null distribution of Loly's invariant)**.** *For a uniformly random arrangement, $I(S)$ of Corollary 3.5 satisfies*

$$\mathbb{E}[I(S)] = \frac{n^2(n^4-1)}{12}, \qquad \mathrm{Var}\big(I(S)\big) = N^2\sigma_n^2\,\mathrm{Var}(X^2 + Y^2) = \frac{n^4(n^4-1)(n^2-4)}{1080}. \tag{16}$$

The mean equals Loly's value for every distribution of arrangements invariant under relabelling of the values; magic and semi-magic squares sit exactly at the mean, with no fluctuation.

Proof. By (10), $I(S) - \frac{n^2(n^4-1)}{12} = N\,T_{X^2+Y^2}$, so Proposition 4.1 gives the mean and $\mathrm{Var}(I) = N^2\sigma_n^2\mathrm{Var}(X^2+Y^2)$. X and Y are independent and uniform on n equally spaced points, so $\mathrm{Var}(X^2+Y^2) = 2(\mathbb{E}X^4 - (\mathbb{E}X^2)^2)$ with $\mathbb{E}X^2 = \frac{n^2-1}{12}$ and $\mathbb{E}X^4 = \frac{(n^2-1)(3n^2-7)}{240}$, which gives $\frac{(n^2-1)(n^2-4)}{90}$. □

For $n = 3$, $\mathrm{Var}(I) = 30$ and I takes 29 values from 46 to 74; for $n = 4$, $\mathrm{Var}(I) = 2176/3$ and $\Pr(I = 340) = 13049/450450 \approx 0.029$ (`anc/verify_numbers.py`).

4.2 Variances

Write a random energy as a quadratic form $E = \sum_k T_{\varphi_k}^2 = z^{\mathsf{T}}Az$ with $A = \sum_k a_k a_k^{\mathsf{T}}$, where $a_k = (\varphi_k - \bar{\varphi}_k)/N$. A is symmetric with zero row sums.

Theorem 4.3 (Variance of a quadratic form in a random arrangement)**.** *Let A be a symmetric $N \times N$ matrix with zero row sums, $N \geq 4$, and let z be a uniformly random permutation of the centred values $k - \frac{N+1}{2}$, $1 \leq k \leq N$. Then $\mathbb{E}[z^{\mathsf{T}}Az] = \frac{N(N+1)}{12}\,\mathrm{tr}\,A$ and*

$$\mathrm{Var}(z^{\mathsf{T}}Az) = -\frac{N(N+1)(2N+3)}{360}\,(\mathrm{tr}\,A)^2 + \frac{N(N^2-1)(5N+6)}{360}\,\mathrm{tr}(A^2) - \frac{N^2(N+1)^2}{120}\sum_a A_{aa}^2. \tag{17}$$

Proof (sketch; details in Section C). Expand $\mathbb{E}[(z^{\mathsf{T}}Az)^2] = \sum_{a,b,c,d} A_{ab}A_{cd}\mathbb{E}[z_a z_b z_c z_d]$ according to the equality pattern of (a,b,c,d), a set partition of $\{1,2,3,4\}$, and apply Möbius inversion on the partition lattice, so that each pattern contributes an unrestricted sum. An unrestricted sum in which some index occurs in only one factor $A_{\cdot\cdot}$ is a row sum of A, hence 0. Only the four partitions without singletons survive, 12|34, 13|24, 14|23 and 1234, contributing $(\mathrm{tr}\,A)^2$, $\mathrm{tr}\,A^2$, $\mathrm{tr}\,A^2$ and $\sum_a A_{aa}^2$, with coefficients given by the joint moments of a random permutation of the centred values. □

The identity is a special case of the classical moment formulas for quadratic assignment statistics [22, 16]. For the energies the three invariants are explicit; this is where the parity of n enters, through the centre cell shared by the two diagonals when n is odd.

Theorem 4.4 (Variances of the energies)**.** *For every $n \geq 3$,*

$$\mathrm{Var}(E_{\mathrm{low}}) = \begin{cases} \dfrac{(n-1)(n^2+1)\,P_{\mathrm{o}}(n)}{129600\,n^6} & n \text{ odd}, \\ \dfrac{(n^2+1)\,P_{\mathrm{e}}(n)}{129600\,n^4} & n \text{ even}, \end{cases} \qquad \mathrm{Var}(E_{\mathrm{full}}) = \begin{cases} \dfrac{(n-1)(n^2+1)\,Q_{\mathrm{o}}(n)}{180\,n^6} & n \text{ odd}, \\ \dfrac{(n-2)(n^2+1)\,Q_{\mathrm{e}}(n)}{180\,n^5} & n \text{ even}, \end{cases}$$

where

$$\begin{aligned} P_{\mathrm{o}} &= 25n^{11} + 25n^{10} - 82n^9 - 82n^8 - 9n^7 - 1209n^6 + 3738n^5 - 5862n^4 \\ &\quad + 2160n^3 - 2880n^2 - 4320n + 4320, \\ P_{\mathrm{e}} &= 25n^{10} - 107n^8 + 73n^6 - 1200n^5 + 4947n^4 - 9600n^3 + 13782n^2 - 13680n + 10080, \\ Q_{\mathrm{o}} &= 5n^6 - 5n^5 - 17n^4 + 11n^3 - 24n^2 + 18n + 6, \qquad Q_{\mathrm{e}} = 5n^5 - 12n^3 + 12n^2 - 23n + 12. \end{aligned}$$

For both parities,

$$\mathrm{Var}(E_{\mathrm{low}}) = \frac{n^8}{5184} - \frac{41\,n^6}{64800} + O(n^4), \qquad \mathrm{Var}(E_{\mathrm{full}}) = \frac{n^3}{36} - \frac{n^2}{18} - \frac{7n}{180} + O(1).$$

Proof. Theorem 4.3 with $N = n^2$ and the invariants (19)–(20) of Section C; the expansions follow. □

At $n = 3$ the formulas give $\mathrm{Var}(E_{\mathrm{low}}) = 1532/2187$ and $\mathrm{Var}(E_{\mathrm{full}}) = 398/2187$, the values obtained by enumerating all 9! arrangements.

4.3 Limit law

Corollary 4.5 (Limit law)**.** *Let $\tau_1^2, \dots, \tau_4^2$ be the variances of $T_X, T_Y, T_{D_\mathrm{m}}, T_{D_\mathrm{a}}$ (the diagonal of $\sigma_n^2\Gamma$). As $n \to \infty$, $\big(T_X/\tau_1, T_Y/\tau_2, T_{D_\mathrm{m}}/\tau_3, T_{D_\mathrm{a}}/\tau_4\big)$ converges in distribution to a standard Gaussian vector in $\mathbb{R}^4$, and*

$$\frac{E_\mathrm{low}}{\mathbb{E}[E_\mathrm{low}]} \xrightarrow{\;d\;} \mathrm{Exp}(1). \tag{18}$$

Proof. Fix $\alpha \in \mathbb{R}^4 \setminus \{0\}$ and let $a = \sum_k \alpha_k a_k/\tau_k$ be the corresponding combination of the four centred, standardised test vectors. Then $\sum_k \alpha_k T_k/\tau_k = \sum_c a_c z_c$ is a linear permutation statistic with doubly centred array $d(c,v) = a_c b_v$, and

$$\left(\frac{\max_{c,v} a_c^2 b_v^2}{N^{-1}\sum_{c,v} a_c^2 b_v^2}\right)^{1/2} = \frac{\max_c |a_c|}{\|a\|_2} \cdot \frac{\max_v |b_v|}{(\sum_v b_v^2/N)^{1/2}};$$

this is the square root of the quotient in (9) (equivalently, of the condition of Hoeffding's Theorem 4). The second factor tends to $\sqrt{3}$. For the first, $\max_c |a_c| \le \sum_k |\alpha_k| \max_c |a_{k,c}|/\tau_k$, and $\max_c |a_{k,c}|/\|a_k\|_2$ is $O(n^{-1})$ for X and Y and $O(n^{-1/2})$ for a diagonal indicator (a centred indicator of n of the N cells); moreover $\|a\|_2^2$ is proportional to $\alpha^\mathsf{T} C_n \alpha$ with C_n the correlation matrix, which tends to the identity because the only nonzero correlation is $\mathrm{corr}(T_{D_\mathrm{m}}, T_{D_\mathrm{a}}) = -1/(n-1)$ for even n. So the quotient tends to 0, condition (9) holds, Hoeffding's Theorem 3 [15] (Section 2.3) gives asymptotic normality of every such combination, and the Cramér–Wold device gives the joint limit $N(0, I_4)$. In $E_\mathrm{low} = \tau_1^2(\tilde T_X^2 + \tilde T_Y^2) + \tau_3^2 \tilde T_{D_\mathrm{m}}^2 + \tau_4^2 \tilde T_{D_\mathrm{a}}^2$ (tildes denoting standardised statistics) the diagonal weights satisfy $\tau_3^2/\tau_1^2 = 12(n-1)/(n^2(n^2-1)) \to 0$, so $E_\mathrm{low}/(2\tau_1^2) \to \tfrac{1}{2}\chi_2^2 = \mathrm{Exp}(1)$; and $\mathbb{E}[E_\mathrm{low}]/(2\tau_1^2) \to 1$. □

Remark 4.6 (Gaussian approximation at finite n). The Gaussian approximation replaces an energy by $\sum_j w_j \chi_{1,j}^2$, with independent χ_1^2 variables and weights w_j the eigenvalues (15); its density can be computed by inversion of the characteristic function [17] (Figure 3). It has the exact mean but variance $2\sum_j w_j^2$, and by Theorem 4.4 the ratio of the exact variance to this value is

$$1 - \frac{2}{n} - \frac{7}{5n^2} + O(n^{-3}) \quad (E_\mathrm{full}), \qquad 1 - \frac{82}{25n^2} + O(n^{-3}) \quad (E_\mathrm{low});$$

the deficit comes from the two negative terms of (17). For E_full the number of statistics grows with n and Corollary 4.5 does not apply. The approximation is poor for small n: at $n = 3$ the Gaussian standard deviation of E_full is about twice the exact one (Table 2).

n	energy	$\mathbb{E}[E]$	$\sqrt{\mathrm{Var}E}$	Gaussian sd	variance ratio
3	E_low	40/27	0.8370	1.171	0.511
3	E_full	40/27	0.4266	0.869	0.241
4	E_low	391/96	3.1284	3.586	0.761
4	E_full	85/32	0.9074	1.349	0.453
5	E_low	234/25	8.0483	8.694	0.857
5	E_full	104/25	1.3966	1.867	0.560
6	E_low	2035/108	17.122	18.007	0.904
6	E_full	1295/216	1.9671	2.459	0.640
7	E_low	5050/147	32.174	33.349	0.931
7	E_full	400/49	2.5697	3.089	0.692
8	E_low	22295/384	55.377	56.888	0.948
8	E_full	1365/128	3.2352	3.777	0.734

Table 2: Exact means and standard deviations of the two energies (Proposition 4.1, Theorem 4.4), the standard deviation of the Gaussian approximation of Remark 4.6, and the ratio of the exact to the Gaussian variance.

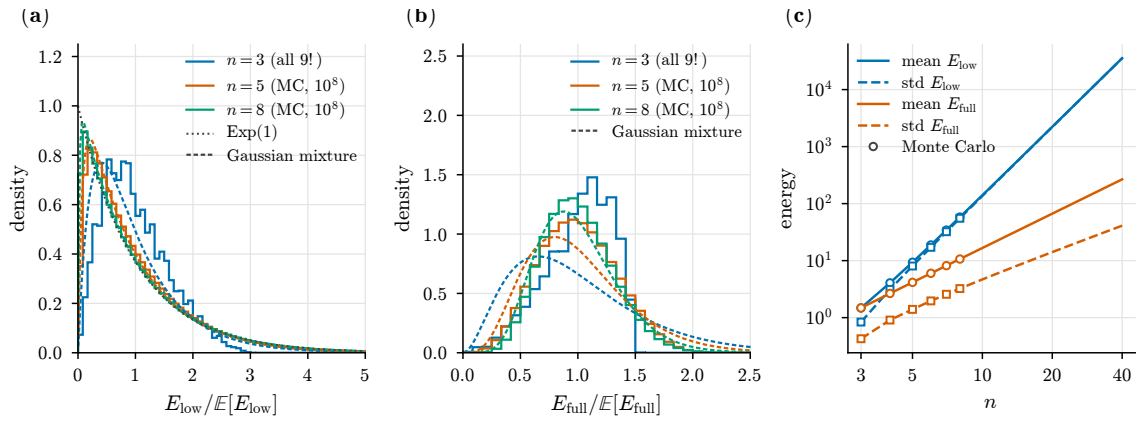


Figure 3: Energy of a uniformly random arrangement. (a) Distribution of $E_{\text{low}}/\mathbb{E}[E_{\text{low}}]$ for $n = 3$ (all 9! arrangements) and $n = 5, 8$ (10^8 Monte Carlo samples each), with the densities of the Gaussian approximation (dashed) and the limit Exp(1) of Corollary 4.5 (dotted). (b) The same for E_{full}, where the Gaussian approximation improves slowly (Remark 4.6). (c) Exact mean (solid) and standard deviation (dashed) of both energies for $3 \le n \le 40$, with Monte Carlo values for $3 \le n \le 8$ (open markers).

5 Computations for small orders

All counts below were computed in exact integer arithmetic; the methods are described in Section A, and every number is produced by a script in the ancillary directory `anc/`.

5.1 The low-order class for $n = 4$

Let L_n be the set of arrangements with $E_{\text{low}} = 0$, i.e. $\text{Cov}(X, Z) = \text{Cov}(Y, Z) = 0$ and both diagonal sums equal to $M(n)$. With cells indexed by (i, j), these are the conditions $U_1 = U_2 = \frac{n-1}{2} \cdot \frac{N(N+1)}{2}$ and $U_3 = U_4 = M(n)$ on the sums $U_1 = \sum_{i,j} j\, s_{ij}$, $U_2 = \sum_{i,j} i\, s_{ij}$, $U_3 = \sum_i s_{ii}$, $U_4 = \sum_i s_{i,n-1-i}$, which are additive over cells.

Proposition 5.1 (The size of L_4)**.** $|L_3| = 8$ *(the magic squares), and*

$$|L_4| = 14{,}639{,}440 = 8 \times 1{,}829{,}930,$$

of which 7,040 *are magic squares. The class* $\{\text{Cov}(X, Z) = \text{Cov}(Y, Z) = 0\}$ *has* 6,697,177,424 *members for* $n = 4$*, and the class with both diagonal sums equal to* 34 *has* 49,932,288,000*.*

Computation. A meet-in-the-middle enumeration over $(U_1, \ldots, U_4)$, splitting the grid into the top two and bottom two rows (Section A); the matching pairs were expanded to obtain the breakdown of Table 3. The same program reproduces the brute-force counts for $n = 3$ (8, 760, 4224, and 72 semi-magic squares) and, with the $2n$ line sums in place of U_1, U_2, the 7,040 magic and 549,504 semi-magic squares of order 4. The diagonal-only count also follows from a direct count: there are 2150 ordered pairs of disjoint 4-subsets of $\{1, \ldots, 16\}$ with sum 34, and $2150 \cdot (4!)^2 \cdot 8! = 49{,}932{,}288{,}000$. □

Since D_4 acts freely on the set of all arrangements (entries are distinct) and preserves L_4, L_4 consists of 1,829,930 orbits of size 8.

Exactly two or three rows with sum 34 are impossible in L_4: the row deviations e_i satisfy $\sum_i e_i = 0$ and $\sum_i i e_i = 0$, which cannot hold with exactly one or two nonzero e_i; the same holds for columns. Thus $7040/14{,}639{,}440 \approx 0.048\%$ of L_4 is magic, and 9,637,312 members (66%) have no row or column with sum 34, among them the arrangement of Remark 2.5.

A local-limit estimate. Under a uniformly random arrangement the vector $(U_1, \ldots, U_4)$ has mean equal to the target vector and covariance matrix Σ_U given by (6): U_1, U_2 are uncorrelated with each other and with the diagonal sums, which have correlation $-1/(n-1)$ for even n and

	columns with sum 34			
rows with sum 34	0	1	4	total
0	9 637 312	2 172 824	100 728	11 910 864
1	2 172 824	429 312	9 336	2 611 472
4	100 728	9 336	**7 040** (magic)	117 104
total	11 910 864	2 611 472	117 104	14 639 440

Table 3: The members of L_4 by the number of rows and of columns with sum $M(4) = 34$; the table is symmetric under transposition.

0 for odd n. The single transpositions change $(U_1, \dots, U_4)$ by vectors whose $\mathbb{Z}$-span is $\mathbb{Z}^4$ (Smith normal form, $n \le 8$). A Gaussian local limit would therefore give $|L_n| \approx N!\,(2\pi)^{-2}\det(\Sigma_U)^{-1/2}$. This estimate is 13.6 for $n = 3$ ($|L_3| = 8$) and 1.82×10^7 for $n = 4$, so $|L_4|$ is 0.80 times the estimate; for the two-condition classes the ratios at $n = 3, 4$ are $0.59, 0.91$ (first moments) and $1.10, 0.96$ (diagonals). For $n = 5, 6$ Monte Carlo (10^{10} and 2×10^{10} samples, 328 and 62 members of L_n) gives $|L_5| \approx 5.1 \times 10^{17}$ ($\pm 6\%$) and $|L_6| \approx 1.2 \times 10^{33}$ ($\pm 13\%$), ratios 0.76 and 0.78 to the estimate. Since $\det(\Sigma_U)^{1/2}$ grows like n^{13}, one expects $|L_n|/N!$ to decay like n^{-13}, and the fraction of magic squares in L_n to tend to 0; local limit theorems for permutation statistics of this kind are studied in [18, 11], and related counts of matrices with prescribed margins in [10].

5.2 Single swaps

Remark 5.2 (Single-swap formula). Let S^{ab} be obtained from S by exchanging the entries of cells $a \ne b$, let $d = s_b - s_a$, let $\lambda(a, b)$ be the number of lines containing exactly one of a, b, and $u_S(c) = \sum_{L \ni c} \delta_L(S)$. Expanding (5),

$$N^2\big(E_{\text{full}}(S^{ab}) - E_{\text{full}}(S)\big) = d^2\lambda(a,b) + 2d\big(u_S(a) - u_S(b)\big),$$

the incremental evaluation used in local search [29]. Here $\lambda(a, b) = 2[i \ne i'] + 2[j \ne j'] +$ [a diagonal separates a, b]+[the other diagonal separates a, b] $\in \{2, \dots, 6\}$ for $a = (i, j)$, $b = (i', j')$. For a magic square S, $N^2 E_{\text{full}}(S^{ab}) = d^2\lambda(a, b) \ge 2$, so no two magic squares differ by a transposition, and the values lie in the set $\{d^2\lambda : 1 \le d \le N - 1,\ 2 \le \lambda \le 6\}$. The smallest value $g_{\min}(S) = \min_{a \ne b} d^2\lambda(a, b)$ satisfies $2 \le g_{\min}(S) \le 6$: the value 2 occurs if and only if two consecutive values lie in a common row or column and off both diagonals, and consecutive values give at most 6.

Over the $880 \cdot \binom{16}{2} = 105{,}600$ swaps of the 880 magic squares of order 4 up to D_4, the gap $d^2\lambda$ takes 70 values from 2 to 980, and $g_{\min}(S)$ is 2 for 432 squares, 3 for 420 and 4 for 28 (Figure 4); it is 3 for the magic squares of order 3. For $n = 3$, besides the eight magic squares, E_{full} has eight non-magic local minima with respect to swaps, all with $N^2 E_{\text{full}} = 2$; E_{low} has 344 local minima, of which 336 are non-magic. Steepest swap descent on E_{full} from a random arrangement reaches a magic square in 19.5% of runs for $n = 4$, 1.2% for $n = 5$ and 0.8% for $n = 6$ (`anc/algorithms.py`).

6 Open questions

The main results are the two-sided bound $(\sqrt{2} - o(1))\sqrt{n} \le k^*(n) \le (2\sqrt{\pi e} + o(1))\sqrt{n \log n}$ for the permutation threshold, with exact values for $n \le 40$, and exact second moments of the line-sum energies of a random arrangement. The following questions remain.

1. *The order of $k^*(n)$.* Is $k^*(n) \asymp \sqrt{n}$? The upper bound uses only that γ is a nonzero integer vector with $\|\gamma\|_1 \le n^4/4$, and for such vectors no bound of smaller order than $\sqrt{n}$ is possible (Section 3.3); the factor $\sqrt{\log n}$ corresponds to the gap left open in [8] for polynomials with coefficients in $\{-1, 0, 1\}$. For $14 \le n \le 40$ the column-feasibility inequalities (11) were the only obstruction (Section 3.5). Is this true for all n, and is $k^*(n)$ nondecreasing?

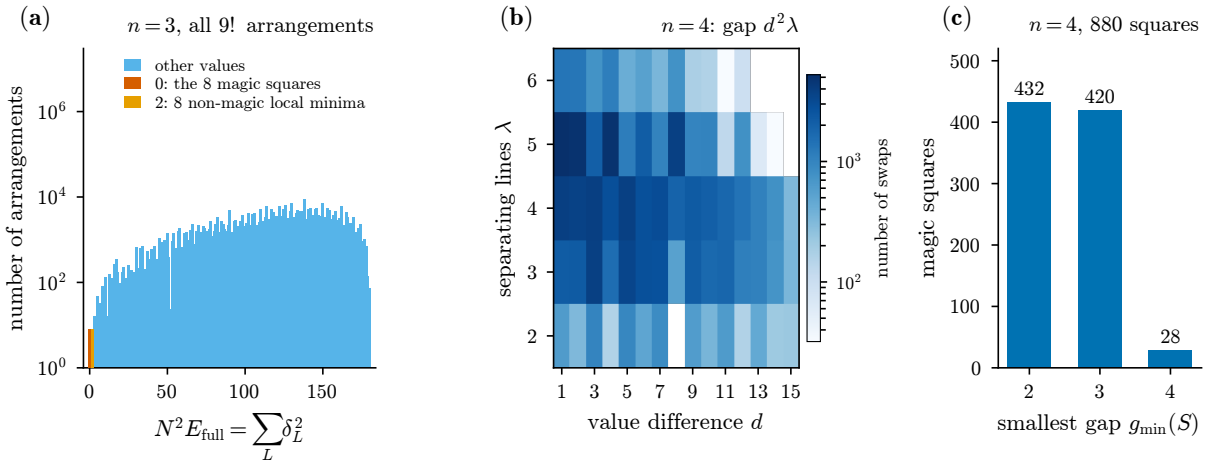


Figure 4: Swap gaps. (a) For $n = 3$, the number of arrangements with each value of $N^2 E_{\text{full}} = \sum_L \delta_L^2$; the eight magic squares are at 0 and the eight non-magic local minima at 2. (b) For $n = 4$, the number of single swaps of the 880 magic squares (105,600 in total) by value difference d and number λ of separating lines; the gap is $d^2\lambda$. (c) The smallest gap $g_{\min}(S)$ over the 880 squares.

2. *The size of the low-order class.* Is $|L_n| \sim N!\,(2\pi)^{-2}\det(\Sigma_U)^{-1/2}\,c_n$ with c_n bounded away from 0 and ∞, i.e. does a local limit theorem hold for the four statistics $(U_1, \ldots, U_4)$ of Section 5.1? The ratio c_n is 0.59, 0.80 for $n = 3, 4$ and about 0.76, 0.78 for $n = 5, 6$.

3. *Unit-vector balance.* Is there a magic square of some order whose ring imbalance $|\sum_c z_c u_c|$ (Section 3.7) vanishes? None exists for $n = 3$ or $n = 4$.

4. *Other families.* The orthogonality formulation applies to pandiagonal and Franklin squares [20], with W replaced by the span of the relevant line indicators. The corresponding thresholds and random energies have not been worked out.

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A Computational methods

All computations were run on an Apple M4 Max under macOS with Python 3.9 (NumPy, SciPy, SymPy) and C compiled with Apple clang; uniformly random arrangements were generated by Fisher–Yates shuffles. The scripts and data are in `anc/`, with a `README` listing the commands that regenerate every table and figure.

The permutation threshold. `anc/kstar_verify.py` checks in exact integer arithmetic that each witness arrangement ($5 \le n \le 40$) is a permutation of $1, \dots, N$ with all row sums $M(n)$, and computes the order of vanishing of P_γ at 1 from the binomial moments $\sum_j \binom{j}{r}\gamma_j$. For the upper bounds it reduces the basis $\{t^i(t-1)^{k+1}\}$ of $\Lambda_{n,k}$ by LLL (exact integer arithmetic) and enumerates all lattice vectors with $\|\gamma\|_2^2 \le n^5(n^2-1)/12$ by the Fincke–Pohst method with exact rational Gram–Schmidt coefficients, testing (11) for every p. Since $\Lambda_{n,k+1} \subset \Lambda_{n,k}$, feasibility is monotone in k, and the least level without a feasible vector is the upper bound. The same script checks the rank-2 lattices $\Lambda_{n,n-3}$ for $14 \le n \le 400$. The witnesses were found with `anc/realise.py` and `anc/anneal2.c`, starting from the squares of Lemma 3.12 (`anc/constructions.py`; an independent implementation of the construction is `anc/twin_squares.py`).

Variances. `anc/variance_proof.py` derives (17) symbolically by Möbius inversion over the 15 set partitions of $\{1,2,3,4\}$, checks it by enumerating all permutations for random matrices with $N \le 7$, computes the invariants (19)–(20) symbolically and checks them against the explicit matrices for $3 \le n \le 25$, and compares the result with the enumeration of all 9! arrangements for $n = 3$. Monte Carlo with 10^8 arrangements per order ($3 \le n \le 8$) agrees with the exact means and standard deviations to within 5×10^{-4} relative error (`anc/mc_results.json`). The densities in Figure 3 were computed by numerical inversion of the characteristic function [17].

Meet-in-the-middle enumeration of L_4. The 16 cells are split into the top two rows A and the bottom two rows B. For each of the $\binom{16}{8} = 12{,}870$ value sets V and each of the 8! bijections $V \to A$ the partial vector $(U_1, \dots, U_4)|_A$ is packed into an integer key; for the complementary set and each bijection $V^c \to B$ the key of the target minus the partial vector is formed; the two key lists are sorted and merged, and the sum over matching keys of the products of multiplicities is $|L_4|$. This touches about 1.04×10^9 partial vectors instead of $16! \approx 2.1 \times 10^{13}$ arrangements (`anc/lowmode_enum.c`).

`anc/verify_numbers.py` and `anc/balance_check.py` recompute the remaining numbers (the former in exact arithmetic).

B Squares with many twin rows

Proof of Lemma 3.12(1). Give the value v the weight $|2v-(N+1)|$, an odd number in $\{1, 3, \dots, N-1\}$; each weight w belongs to exactly the two complementary values $(N+1 \pm w)/2$. Index the weights by $x = (w-1)/2 \in \{0, \dots, 2m^2-1\}$, $w_x = 2x+1$. It suffices to partition the indices into m sets $R_0, \dots, R_{m-1}$ of size n and choose signs $\sigma_x \in \{\pm 1\}$ with $\sum_{x \in R_r} \sigma_x w_x = 0$ for every r: then row r receives the values $(N+1+\sigma_x w_x)/2$, $x \in R_r$, in any order, and row $m+r$ their complements in the same columns. Every value occurs once, row r sums to $n\frac{N+1}{2} = M(n)$, row $m+r$ to $n(N+1) - M(n) = M(n)$, and every column to $m(N+1) = M(n)$. A *domino* $\{x, x+1\}$ with signs $(+,-)$ or $(-,+)$ contributes ∓ 2. If m is even, give each row m of the dominoes $\{2t, 2t+1\}$ with alternating sign patterns. If $m \ge 7$ is odd, give row r the four indices $4r, \dots, 4r+3$ with signs $(+,+,+,-)$, contributing $(8r+1)+(8r+3)+(8r+5)-(8r+7) = 16r+2$; the two indices $\alpha_r = 5m-1-r$ and $\beta_r = 5m+7r$ with signs $(+,-)$, contributing $2(\alpha_r - \beta_r) = -16r-2$; and $m-3$ dominoes with alternating sign patterns. The quadruples fill $[0, 4m-1]$ and the α_r fill $[4m, 5m-1]$; consecutive β_r leave gaps of exactly 6 indices, and the remaining interval $[12m-6,\ 2m^2-1]$ has even length $2m^2-12m+6 \ge 0$ because $m \ge 7$. So the unused indices form $3(m-1) + (m^2-6m+3) = m(m-3)$ dominoes, as required. □

Proof of Lemma 3.12(2). Start with a 7×7 magic square and border it successively to the orders $9, 11, \dots, n$ by the following step, which turns a semi-magic square T of order $n-2$ into one of order $n = 2m+1$ ($m \ge 4$) whose top and bottom rows are twins on the inner columns. Let $N = n^2$, $c = (N+1)/2$, $A = 4m$, $B = 4m-2$. The inner block is $T + (2n-2)$, which uses the values $2n-1, \dots, N-2n+2$; each inner row and column then sums to $M(n-2) + (n-2)(2n-2) = (n-2)(N+1)/2$. The border uses the $4m = 2n-2$ complementary pairs $\{p, N+1-p\}$, $1 \le p \le 4m$. Put $s_{00} = N+1-A$, $s_{n-1,n-1} = A$, $s_{0,n-1} = B$, $s_{n-1,0} = N+1-B$. Let $P = [2m, 4m] \setminus \{A, B\}$ and $Q = [1, 2m-1]$, both of size $2m-1$, so $\sum P - \sum Q = 4m^2 - 4m + 2$. Choose an m-subset T_+ of $[2m, 4m-3] \subset P$ and an $(m-1)$-subset T_- of Q with $\sum T_+ - \sum T_- = c - A + B = 2m^2 + 2m - 1$. This is possible: the k-subsets of an interval of integers have as sums all integers between the smallest and the largest such sum (raise one element by 1 at a time), so $\sum T_+ - \sum T_-$ takes every integer in $[m^2+m,\ 3m^2-2m]$, which contains $2m^2+2m-1$ for $m \ge 4$. The inner top row consists of the values $N+1-p$ ($p \in T_+$) and p ($p \in T_-$); the inner bottom row of their complements in the same columns; the inner left column of $N+1-p$ ($p \in P \setminus T_+$) and p ($p \in Q \setminus T_-$); and the inner right column of their complements in the same rows. The top row sums to $(m+1)(N+1) - A + B - \sum T_+ + \sum T_- = (2m+1)c = M(n)$, and the left column to $(m+1)(N+1) - A - B - \sum(P \setminus T_+) + \sum(Q \setminus T_-) = M(n)$,

because $\sum(P \setminus T_+) - \sum(Q \setminus T_-) = 4m^2 - 4m + 2 - (2m^2 + 2m - 1) = c - A - B$. The bottom row and right column then sum to $n(N+1) - M(n) = M(n)$, each inner row and column to $N+1+(n-2)(N+1)/2 = M(n)$, and every value $1, \dots, N$ occurs once. Bordering shifts the inner square by a constant s, and $(a+s)+(b+s) = N+1$ whenever $a+b$ equals the previous $N+1$, so earlier twin relations persist; this gives the twin rows $\ell, n-1-\ell$ on the columns $\ell+1, \dots, n-2-\ell$ for $0 \le \ell \le (n-9)/2$. □

C Derivation of the variance formulas

Joint moments. Let b be a vector with $\sum_k b_k = 0$, $p_r = \sum_k b_k^r$, and z a uniformly random permutation of b. For distinct cells, $\mathbb{E}[z_a^4] = p_4/N$, and with $(N)_k = N(N-1)\cdots(N-k+1)$,

$$\mathbb{E}[z_a^3 z_b] = \frac{-p_4}{(N)_2}, \quad \mathbb{E}[z_a^2 z_b^2] = \frac{p_2^2 - p_4}{(N)_2}, \quad \mathbb{E}[z_a^2 z_b z_c] = \frac{2p_4 - p_2^2}{(N)_3}, \quad \mathbb{E}[z_a z_b z_c z_d] = \frac{3p_2^2 - 6p_4}{(N)_4},$$

by Möbius inversion of the power sums over set partitions. Write m_π for $\mathbb{E}[z_a z_b z_c z_d]$ when the equality pattern of (a,b,c,d) is exactly the partition π of the positions $\{1,2,3,4\}$.

Proof of Theorem 4.3. The mean: $\mathbb{E}[z^\mathsf{T} A z] = \frac{p_2}{N} \operatorname{tr} A - \frac{p_2}{N(N-1)} \sum_{a \ne b} A_{ab}$ and $\sum_{a \ne b} A_{ab} = -\operatorname{tr} A$. For the second moment let $S^{=}(\pi)$ be the sum of $A_{ab}A_{cd}$ over the tuples with pattern exactly π and $S^{\ge}(\sigma) = \sum_{\pi \ge \sigma} S^{=}(\pi)$ the sum over tuples satisfying at least the equalities of σ; $S^{\ge}(\sigma)$ is an unrestricted sum. Möbius inversion gives $\mathbb{E}[(z^\mathsf{T} A z)^2] = \sum_\pi m_\pi S^{=}(\pi) = \sum_\sigma c(\sigma) S^{\ge}(\sigma)$ with $c(\sigma) = \sum_{\pi \le \sigma} \mu(\pi, \sigma) m_\pi$. If σ has a singleton block, the corresponding index occurs in a single factor of $A_{ab}A_{cd}$ and summing over it gives a row sum, so $S^{\ge}(\sigma) = 0$. The remaining partitions are $12|34$, $13|24$, $14|23$ and 1234, with $S^{\ge}$ equal to $(\operatorname{tr} A)^2$, $\operatorname{tr} A^2$, $\operatorname{tr} A^2$, $\sum_a A_{aa}^2$, and

$$c(12|34) = c(13|24) = c(14|23) = m_{22} - 2m_{211} + m_{1111},$$
$$c(1234) = m_4 - 4m_{31} - 3m_{22} + 12m_{211} - 6m_{1111}.$$

For the centred values $1, \dots, N$, $p_2 = N(N^2-1)/12$ and $p_4 = N(N^2-1)(3N^2-7)/240$, and these coefficients simplify to $N(N^2-1)(5N+6)/720$ and $-N^2(N+1)^2/120$; subtracting $\left(\frac{N(N+1)}{12} \operatorname{tr} A\right)^2$ gives (17).

The invariants. With $\Gamma_{kl} = \operatorname{Cov}(\varphi_k, \varphi_l)$ and $h_c = \sum_k a_{k,c}^2$,

$$\operatorname{tr} A = \frac{1}{N} \operatorname{tr} \Gamma, \qquad \operatorname{tr} A^2 = \frac{1}{N^2} \sum_{k,l} \Gamma_{kl}^2, \qquad \sum_a A_{aa}^2 = \sum_c h_c^2.$$

For a line, $(\mathbf{1}_L(c) - \frac{1}{n})^2 = \frac{1}{n^2} + (1 - \frac{2}{n})\mathbf{1}_L(c)$. Hence $N^2 h_c = x_c^2 + y_c^2 + \frac{2}{n^2} + (1 - \frac{2}{n})(d_\mathrm{m} + d_\mathrm{a})(c)$ for E_low and $N^2 h_c = \frac{2n+2}{n^2} + (1 - \frac{2}{n})(2 + d_\mathrm{m}(c) + d_\mathrm{a}(c))$ for E_full, where $d_\mathrm{m}, d_\mathrm{a}$ are the diagonal indicators. The entries of Γ are given in the proof of Proposition 4.1. The sums needed are $\sum_c x_c^2 = \frac{n^2(n^2-1)}{12}$, $\sum_c x_c^4 = \frac{n^2(n^2-1)(3n^2-7)}{240}$, $\sum_c x_c^2 y_c^2 = \left(\frac{n(n^2-1)}{12}\right)^2$, $\sum_{c \in D} x_c^2 = \sum_{c \in D} y_c^2 = \frac{n(n^2-1)}{12}$ for either diagonal D, $\sum_c (d_\mathrm{m} + d_\mathrm{a}) = 2n$ and $\sum_c (d_\mathrm{m} + d_\mathrm{a})^2 = 2n + 2\varepsilon$, where $\varepsilon = 1$ for odd and 0 for even n is the number of cells on both diagonals. Collecting terms gives the following invariants; where two values are given, the first is for odd and the second for even n. For E_low:

$$\begin{aligned}
\operatorname{tr} A &= \frac{(n-1)(n^3+n^2+12)}{6n^4}, \\
\operatorname{tr} A^2 &= \frac{(n-1)^2(n^6+2n^5+n^4+144)}{72n^8}, \qquad \frac{n^8-2n^6+n^4+144n^2-288n+288}{72n^8}, \\
\sum_a A_{aa}^2 &= \frac{(n-1)(7n^7+7n^6-13n^5+107n^4+240n^2-720n+720)}{180n^{10}}, \\
&\qquad \frac{7n^8-20n^6+120n^5-107n^4+240n^3-1320n^2+2880n-2160}{180n^{10}}.
\end{aligned} \tag{19}$$

For E_{full}:

$$\begin{aligned} \operatorname{tr} A &= \frac{2(n^2-1)}{n^4}, \qquad \operatorname{tr} A^2 = \frac{2(n-1)(n^2+n-1)}{n^8}, \quad \frac{2(n^3-2n+2)}{n^8}, \\ \sum_a A_{aa}^2 &= \frac{2(n-1)(2n-1)(n^2+2n-2)}{n^{10}}, \quad \frac{2(2n^4+n^3-10n^2+12n-6)}{n^{10}}. \end{aligned} \tag{20}$$

D A witness for $k^*(5) = 4$

The arrangement

$$\begin{pmatrix} 1 & 22 & 11 & 24 & 7 \\ 23 & 3 & 12 & 2 & 25 \\ 18 & 6 & 19 & 17 & 5 \\ 4 & 21 & 14 & 10 & 16 \\ 20 & 9 & 15 & 8 & 13 \end{pmatrix}$$

has all row sums $M(5) = 65$ and column sums $66, 61, 71, 61, 66$, i.e. $\gamma = (1, -4, 6, -4, 1)$ and $P_\gamma = (1-t)^4$; it satisfies the degree-3 conditions without being semi-magic. The witnesses for $6 \le n \le 40$ are in `anc/data_witnesses_variances.json` ($n \le 13$) and `anc/kstar_witnesses.json` ($14 \le n \le 40$).